

MOHAN S

B.Tech – Artificial Intelligence and Machine Learning

+91 73973 18722 | mohansithaiya@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

B.Tech student specializing in Artificial Intelligence and Machine Learning, with hands-on experience building computer vision systems, full-stack web applications, and API services. Practical experience with PyTorch, object detection (RT-DETR, YOLO), FastAPI, and the MERN stack, backed by a strong foundation in Data Structures, OOP, DBMS, and Git-based software development.

EDUCATION

Saveetha Engineering College

Chennai, Tamil Nadu

B.Tech in Artificial Intelligence and Machine Learning

2023 – 2027

CGPA: 7.94 / 10

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL

Web & APIs: React, Node.js, Express, FastAPI, REST APIs, HTML, CSS

AI / ML: PyTorch, Computer Vision, Object Detection (RT-DETR, YOLO), OCR, OpenCV, NumPy, Pandas, Matplotlib, scikit-learn

Databases & Tools: MongoDB, SQL, Git, GitHub, Docker, VS Code, Linux, Jupyter Notebook

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

Electrical Equipment Inspection — Object Detection & Reasoning API — github.com/Mohansithaiya/rap-electrical-inspection-api

Python, PyTorch, RT-DETR, FastAPI, Docker

- Fine-tuned RT-DETR-L on 8 custom electrical-equipment and hazard classes, reaching 0.933 mAP50, 0.916 precision, and 0.896 recall on a 568-image held-out test set.
- Built a FastAPI service (/detect, /ask) with a deterministic, rule-based natural-language reasoning layer — no LLM — that answers presence, count, location, and risk questions from detection results.
- Automated the COCO-to-YOLO dataset pipeline with MD5-based duplicate and train/test leakage detection; containerized the service with Docker and covered it with pytest unit tests.

NAMMANADU — Civic Complaint Management Platform — github.com/Mohansithaiya/NAMMANADU-Platform

React, Node.js, Express, MongoDB, JWT

- Built a full-stack MERN platform for citizens to file, track, and search civic complaints, with JWT access/refresh-token authentication, bcrypt password hashing, and role-based access control.
- Implemented a complaint API with server-side pagination, search, and status/category filtering, plus Cloudinary-backed photo-evidence uploads validated by file signature rather than declared file type.
- Designed the React 19 / Vite citizen dashboard for filing complaints and tracking their status end to end.

Cat-Dog-Panda Image Classifier — github.com/Mohansithaiya/cat-dog-panda-classifier

Python, PyTorch, ResNet18, Streamlit

- Built an image classifier using transfer learning on a pretrained, frozen ResNet18 backbone with a custom classification head, dropout regularization, and data augmentation.
- Built a Streamlit web app for interactive image upload and real-time prediction.

EXPERIENCE

Zybeak Technologies Pvt. Ltd.

Chennai, Tamil Nadu

Cybersecurity & Ethical Hacking Intern

Jan 2025

- Performed vulnerability assessment and security analysis on systems and web applications using cybersecurity tools, assisting with penetration-testing and troubleshooting activities.

CERTIFICATIONS

IBM: Introduction to Machine Learning, Regression & Classification, Prompt Engineering

JPMorgan Chase & Co. (Forage): Software Engineering Job Simulation | Great Learning: Python Fundamentals

University of Colorado Boulder: Database Management Essentials | AWS Skill Builder: Cloud Foundations

University of Michigan: Django for Everybody | NVIDIA: Computer Vision for Industrial Inspection

ACHIEVEMENTS

Finalist: Yukthi CTF 2025 | Finalist: DEFY 2026, VIT | Hackathons: Smart India Hackathon, IBM Datathon, IIT Madras & VIT Hackathons